

## A nonlinear system

$$y = x^2 \quad y = x + 2$$

**Which equation makes the system nonlinear?**

## Checking an ordered pair

$$y = x^2 \quad x + y = 6$$

**Does (2, 4) satisfy both equations?**

## A line and a parabola

$$y = x^2 \quad y = x + 2$$

**What are all real intersection points?**

## A tangent intersection

$$y = x^2 \quad y = 2x - 1$$

**How many real intersection points does this system have?**

## No real intersections

$$y = x^2 + 2 \quad y = -1$$

**Does this system have any real solution?**

## Substitution and factoring

$$y = x^2 - 4 \quad y = 2x - 1$$

**What are all real solutions?**

## A circle and a line

$$x^2 + y^2 = 25 \quad y = x + 1$$

**What are all real solutions?**

## A circle and an axis-parallel line

$$x^2 + y^2 = 25 \quad x = 3$$

**What are all real solutions?**

## Eliminating squared terms

$$x^2 + y^2 = 25 \quad x^2 - y^2 = 7$$

**What are all real solutions using elimination?**

## Two intersecting circles

$$x^2 + y^2 = 25 \quad (x - 6)^2 + y^2 = 25$$

**What are all real intersection points?**

## Two nonlinear curves

$$y = x^2 \quad y = 4 - x^2$$

**What are all real intersections?**

## Preserving both signs

$$x^2 + y^2 = 20 \quad y^2 = 4$$

**How many ordered-pair solutions does the system have?**

## A product and a sum

$$x + y = 7 \quad xy = 10$$

**What are all real ordered-pair solutions?**

## An inequality solution

$$y > x^2 - 1$$

**Does (0, 0) belong to the solution region?**

## Boundary of a region

$$y \leq x^2 + 2$$

**What curve forms the boundary, and is it included?**

## Choosing a side

$$y < x^2 - 4$$

**Which side of the parabola contains the solution region?**

## Inside or outside a circle

$$x^2 + y^2 < 9$$

**What region represents the solution set?**

## A circular boundary

$$x^2 + y^2 \geq 16$$

**Is the circle included, and which side is shaded?**

## A system of inequalities

$$y \geq x^2 \quad y \leq 4$$

**What region satisfies both inequalities?**

## Testing a point in a system

$$y > x^2 \quad y < 3 - x$$

**Does (1, 2) satisfy both inequalities?**

## Intersection boundaries

$$y \geq x^2 \quad y \leq x + 2$$

**Where do the two boundary curves intersect?**

## A mixed-boundary region

$$y > x^2 \quad y \leq x + 2$$

**What shaded region shows the included and excluded boundaries?**

## A circle and a half-plane

$$x^2 + y^2 \leq 16 \quad y \geq x$$

**What common shaded region satisfies the system?**

## A feasible region

The variables  $x$  and  $y$  are constrained quantities; also require  $x \geq 0$  and  $y \geq 0$ .

$$y \geq x^2 \quad y \leq 9$$

**What feasible region also satisfies the nonnegative-coordinate constraints?**